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Department of Computer Science

Aviation Engine Health Monitoring via Machine Learning

**NGAFID: Anomaly Detection & Fault Classification on a Real Cessna-172
Fleet**

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Abstract

This report presents a machine learning study on aviation engine health monitoring using the National General Aviation Flight Information Database (NGAFID), a real-world dataset of 11,446 flights from a Cessna-172 flight school fleet labeled with 19 fault categories and correlated with over 2,000 unscheduled maintenance events. We formulate the problem as a two-stage pipeline: binary Anomaly Detection (AD) followed by 19-class Fault Classification (FC). Multiple classical models (Logistic Regression, SVM, XGBoost, Random Forest) are trained on flight-level statistical feature representations and benchmarked against deep learning models from the literature, including ConvTokMHSA, MMK Net, and the LMSD framework. Our best classical system achieves $F1 = 0.691$ on AD (Logistic Regression, 184 features) and $F1 = 0.420$ on FC (Oracle $K=4$ hierarchical grouping), compared to the DL benchmarks of 0.799 and 0.623 respectively. The analysis shows that global statistical features are competitive for binary detection, but a structural 40-point gap persists on FC because fault signatures concentrate in the 12th–18th principal components, contributing less than 0.1% of total variance and remaining invisible to global statistics. The end-to-end pipeline achieves $F1 = 0.184$, limited primarily by Stage 2 fault classification errors (40.6% of all errors).

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1. Introduction

1.1 Motivation

General aviation in the United States operates under a predominantly *calendar-based* maintenance paradigm: aircraft are inspected at fixed intervals of hours, days, or calendar cycles regardless of the actual condition of their components. This reactive approach systematically fails to detect emerging faults between inspection windows, creating safety risks and generating costly unscheduled maintenance events that grounding aircraft without warning.

Modern aircraft are equipped with flight data recorders (FDR) and engine control units (ECU) that continuously log dozens of sensor channels at 1 Hz or higher. This operational data, accumulated across thousands of flight hours, encodes the health history of every critical subsystem. Machine learning offers a principled path from raw telemetry to *condition-based* maintenance: instead of waiting for a calendar date, a model can flag anomalous flights and identify the likely fault category from the sensor profile alone, enabling targeted interventions before failure occurs.

1.2 Dataset and Scope

This project uses the **National General Aviation Flight Information Database (NGAFID)** [1], a publicly available benchmark comprising over 28,000 real flights from a Cessna-172 fleet operated by an active flight school. NGAFID is unique among aviation health datasets in that it provides (i) authentic multi-sensor telemetry under real operational conditions, (ii) fault labels derived from verified unscheduled maintenance records spanning 36 distinct categories, and (iii) a standardized 5-fold cross-validation split designed to prevent data leakage between flights of the same aircraft [2].

The analysis focuses on the *benchmark 2-day subset*: 11,446 flights across 19 fault categories, preprocessed to fixed length via cubic spline interpolation to 2,048 time steps per flight.

1.3 Research Questions

This report investigates two core questions:

1. **Anomaly Detection (AD):** Can a classical ML classifier, trained on statistical features extracted from the full flight profile, reliably distinguish healthy flights from anomalous ones?
2. **Fault Classification (FC):** Given that a flight is anomalous, can the same approach identify which of 19 fault types is responsible?

Both tasks are then composed into a single **End-to-End (E2E) diagnosis pipeline** that assigns one of 20 labels—*healthy* or one of 19 fault types—to any incoming flight.

1.4 Hypothesis

We hypothesize that:

- For **AD**, global statistical features (mean, standard deviation, slope, etc. over the full flight) are sufficient to characterize the overall health state of the aircraft, making classical models competitive with deep learning.
- For **FC**, fault signatures manifest as *local micro-patterns* in specific temporal segments of the flight profile, contributing less than 0.1% of total variance. Global statistics discard this information, creating a structural performance gap versus temporal deep learning models that operate directly on the raw time series.

1.5 Contributions

The main contributions of this work are:

1. A systematic evaluation of five classical ML models across multiple feature representations (Repr. B through Repr. F₊₊) on the NGAFID benchmark subset, with direct comparison against published deep learning results [3].
2. A feature engineering analysis that identifies cross-physical inter-cylinder features (CHT/EGT spreads and ratios) as the most informative additions beyond global per-sensor statistics.
3. A hierarchical fault classification experiment (Oracle $K = 4$) that raises the classical ceiling from $F1 = 0.121$ to $F1 = 0.420$ through physics-based fault grouping, closing over half the gap to the DL best (MMK Net, $F1 = 0.623$) without any temporal modeling.
4. An end-to-end error decomposition showing that 40.6% of E2E errors originate in Stage 2 (wrong fault type), identifying fault classification as the bottleneck for future improvement.

1.6 Report Organization

The remainder of this report is organized as follows. Section 2 describes the NGAFID dataset, sensor space, class distribution, and key data challenges. Section 3 presents the feature engineering pipeline and the hierarchy of representations from global statistics to temporal shape. Section 4 details the two-stage diagnosis pipeline, the classical and deep learning models, and the evaluation protocol. Section 5 reports and analyzes results for AD, FC, and E2E across all representations, with comparison to DL benchmarks.

2. Dataset Description

2.1 NGAFID Overview

The National General Aviation Flight Information Database (NGAFID) is a real-world benchmark for aircraft health diagnosis research. It was collected from a Cessna-172 fleet operated by an active flight school and correlates 23-dimensional sensor measurements with verified unscheduled maintenance records. Table 1 summarizes the key characteristics of the dataset.

Table 1: NGAFID dataset overview.

Property	Value
Aircraft type	Cessna-172
Operator	Real flight school (anonymized)
Total flights	>28,000 (31,000 flight hours)
Sensor channels	23 @ 1 Hz
Maintenance categories	36
Unscheduled events	>2,000
Preprocessing	Forward-fill for NaN; cubic spline → 2,048 pts
<i>Benchmark subset (2-day) — used in this work</i>	
Total flights	11,446
Healthy flights	5,845 (51.1%)
Anomalous flights	5,601 (48.9%)
Fault classes (FC)	19
Cross-validation	5-fold stratified, physically isolated

NGAFID provides two dataset configurations: the *benchmark subset* (19 classes, 11,446 flights) used for controlled evaluation, and the *overall dataset* (36 classes, 28,935 flights) for open-environment assessment. This study uses the benchmark subset exclusively.

2.2 Sensor Space

The 23 sensor channels cover five critical aircraft subsystems. Table 2 lists all channels grouped by subsystem.

Table 2: 23-dimensional sensor space organized by subsystem.

Subsystem	Channels	Count
Electrical	volt1, volt2, amp1, amp2	4
Fuel	FQtyL, FQtyR	2
Engine	E1 FFLOW, E1 OilT, E1 OilP, E1 RPM	4
Cylinders (CHT)	E1 CHT1, E1 CHT2, E1 CHT3, E1 CHT4	4
Cylinders (EGT)	E1 EGT1, E1 EGT2, E1 EGT3, E1 EGT4	4
Flight state	IAS, VSpd, AltMSL, NormAc	4
Environment	OAT	1
Total		23

All 23 channels are sampled at 1 Hz. After preprocessing, each flight is represented as a matrix $X \in \mathbb{R}^{2048 \times 23}$, where rows correspond to time steps and columns to sensor channels.

2.3 Flight Duration Statistics

Raw flight lengths vary considerably across the dataset, reflecting the diverse operational modes of a flight school (student training, check rides, cross-country flights). After interpolation to 2,048 fixed time steps, the original durations are preserved as metadata:

- **Minimum:** 605 s (≈ 10 min)
- **Median:** 5,282 s (≈ 88 min)
- **Maximum:** 20,679 s (≈ 345 min)

2.4 Class Distribution and Fault Taxonomy

The 19 fault categories in the benchmark subset represent distinct maintenance findings, each linked to a specific structural or functional component of the aircraft. Table 3 lists all classes in descending order of flight count.

Table 3: Fault class distribution in the benchmark subset (5,601 anomalous flights).

ID	Fault Category	Flights	%
14	Intake gasket leak / damage	2,097	37.4
18	Rocker cover leak / loose / damage	1,024	18.3
2	Baffle crack / damage / loose / miss	304	5.4
4	Intake tube / bolt / seal / boot loose or damage	269	4.8
1	Baffle plug need repair / replace	254	4.5
3	Baffle screw missing / loose	211	3.8
6	Baffle seal loose / damage	197	3.5
8	Engine failure / fire / time out	161	2.9
5	Baffle tie / tie rod loose or damage	144	2.6
10	Cylinder compression issue	143	2.6
11	Engine run rough	141	2.5
7	Cylinder crack / fail / need repair	108	1.9
13	Engine / propeller overspeed or damage	94	1.7
9	Engine idle / RPM issue	93	1.7
15	Engine need repair / reinstall / clean	80	1.4
12	Engine seal / tube / bolt loose or damage	76	1.4
16	Oil cooler need maintenance	75	1.3
17	Pilot / in-flight noticed issue	75	1.3
0	Baffle rivet loose / missing / damage	55	1.0
—	Total anomalous	5,601	100.0

The imbalance ratio between the largest class (intake gasket, $n = 2,097$) and the smallest class (baffle rivet, $n = 55$) is **38.1:1**, creating an extreme long-tailed distribution that poses a significant challenge for minority class recognition.

2.5 Key Data Challenges

NGAFID exhibits three fundamental characteristics that distinguish it from synthetic benchmarks and directly constrain the achievable performance of any classification approach [2, 3].

2.5.1 Target Component Sparsity

When the feature matrix $\mathbf{X}$ is projected into principal component space, fault-discriminative information concentrates in components 12–18, which together account for **less than 0.1% of total variance**. The dominant components (PCs 1–11) capture operational variability common to both healthy and anomalous flights—altitude profile, throttle inputs, ambient temperature effects—and actively *mask* the fault signatures

that classifiers need to detect.

Empirically, the Repr. B feature matrix (184 features) requires **51 principal components** to explain 95% of its variance. This high intrinsic dimensionality relative to the number of fault-informative components explains why deep models with local receptive fields outperform global statistical methods on FC.

2.5.2 Multi-Stage Coupling

Each flight traverses multiple operationally heterogeneous phases: ground taxi, takeoff and climb, cruise, descent and approach, and ground rollout. These phases have statistically distinct dynamics for virtually every sensor channel—RPM, fuel flow, CHT, and altitude all follow phase-specific profiles. This heterogeneity creates strong temporal dependencies that prohibit phase-wise isolation; a feature vector that captures only one flight phase misses the fault’s temporal context. Conversely, a global feature vector aggregates across all phases, diluting phase-specific anomalies.

2.5.3 Dual Imbalance

NGAFID exhibits imbalance at two levels simultaneously:

- **Quantitative (AD level):** The benchmark subset is nearly balanced between healthy (51.1%) and anomalous (48.9%) flights. However, the *overall* NGAFID dataset has a substantial majority of healthy flights, reflecting real fleet operations where chronic wear dominates over acute failures.
- **Structural (FC level):** Among the 19 fault classes, the distribution is severely long-tailed with an imbalance ratio of 38.1:1. The two head classes alone (intake gasket + rocker cover) account for 55.7% of all anomalous flights. This reflects *objective fleet reality*—some faults are inherently more prevalent—rather than sampling bias.

2.6 Cross-Validation Protocol

All experiments use a **5-fold stratified cross-validation** scheme that maintains class proportions within each fold while enforcing strict physical isolation: no two flights from the same physical aircraft appear in both the training and test partitions of the same fold. This prevents optimistic performance estimates that would arise from data leakage between temporally correlated flights.

Fold composition for the AD task is shown in Table 4.

Table 4: 5-fold CV composition for the AD task.

Fold	Healthy	Anomalous
0	1,167	1,123
1	1,123	1,166
2	1,162	1,127
3	1,192	1,097
4	1,201	1,088
Total	5,845	5,601

Feature normalization (z-score standardization) is applied within each fold, with mean and standard deviation estimated exclusively from the training split and applied to the test split to prevent test-set contamination.

3. Feature Engineering

A central challenge in applying classical ML to variable-length time series is the extraction of a fixed-size feature vector that captures diagnostically relevant information while discarding operational noise. This section describes the hierarchy of representations developed for NGAFID, from a simple global statistics baseline to temporal shape descriptors.

3.1 Representation B: Global Per-Sensor Statistics

The foundational representation compresses each flight's $\mathbf{X} \in \mathbb{R}^{2048 \times 23}$ into a fixed vector by computing eight aggregate statistics per sensor channel:

$$\phi(\mathbf{x}_j) = [\mu_j, \sigma_j, x_j^{\min}, x_j^{\max}, r_j, \text{skew}_j, \text{kurt}_j, \hat{\beta}_j] \quad (1)$$

where $\mathbf{x}_j \in \mathbb{R}^{2048}$ is the time series for sensor j , μ_j is the mean, σ_j the standard deviation, $r_j = x_j^{\max} - x_j^{\min}$ the range, and $\hat{\beta}_j$ the ordinary least-squares slope of a linear trend fit to the time series (computed via the closed-form formula $\hat{\beta} = \frac{\sum_i (t - \bar{t})(x_i - \mu)}{\sum_i (t - \bar{t})^2}$ for efficiency).

Concatenating these eight statistics over all 23 channels yields:

$$\mathbf{f}^{(B)} \in \mathbb{R}^{184}, \quad 184 = 23 \times 8 \quad (2)$$

Missing values. Raw features contain 1,164 NaN values (across the full 11,446-flight dataset), arising from sensor drop-outs on individual flights. These are imputed with the column-wise median of the training fold, reducing NaN count to zero before model fitting.

Build time. Computing Repr. B for 11,446 flights takes approximately 185 s on a standard laptop. The linear slope is computed using a fast closed-form expression (approximately 5× faster than `numpy.polyfit`).

PCA analysis. When Repr. B is projected into principal component space, **51 components** are needed to explain 95% of the variance. This indicates a high-dimensional structure with no single dominant direction, and confirms that fault signatures—concentrated in PCs 12–18 at <0.1% of total variance—are effectively buried by the operational variability captured in the top components.

3.2 Representation B_{seg}: Temporal Segmentation

Repr. B discards all temporal structure: a mean of 3,500 RPM is the same whether the engine stayed at that value throughout the flight or ramped up and then down. To reintroduce *temporal ordering* while remaining within the classical ML framework, each flight is divided into N equal-length segments and Repr. B statistics are computed independently on each segment.

Formally, for a flight of 2,048 steps divided into N segments of $L = 2048/N$ steps each, the segmented

feature vector is:

$$\mathbf{f}^{(B_{\text{seg}})} = [\phi(\mathbf{x}_j^{(1)}), \dots, \phi(\mathbf{x}_j^{(N)})]_{j=1}^{23} \in \mathbb{R}^{23 \times 8 \times N} \quad (3)$$

Table 5 summarizes the feature counts and temporal granularity for each segmentation level.

Table 5: Repr. B_{seg} configurations.

N_{seg}	Features	Temporal resolution	Segment length
1	184	None (global)	2,048 steps
3	552	Thirds of flight	682 steps
5	920	Fifths of flight	409 steps
10	1,840	Tenths of flight	204 steps

The hypothesis is that the *evolution* of sensor statistics across flight thirds or fifths carries diagnostic information beyond the global distribution.

3.3 Representation F_{++} : Cross-Physical Features

Beyond per-sensor statistics, physically meaningful relationships between sensors can provide additional discriminative power. Repr. F_{++} extends Repr. B with a set of *inter-cylinder features*:

- **Spread:** Maximum minus minimum across the four CHT (and four EGT) channels at each time step, then summarized with the same eight statistics. A large CHT spread indicates uneven combustion or a failing cylinder—a fault signature invisible to per-channel averages.
- **Coefficient of variation:** $CV = \sigma/\mu$ for CHT and EGT cross-cylinder variability.
- **Cross-cylinder ratios:** Pairwise ratios (e.g., CHT1/CHT2, EGT1/EGT2) under statistical summaries.
- **Extended statistics on extremes:** Slope and range computed on the per-window max and min envelopes, capturing transient events missed by global statistics.

This brings the total feature count to **224 features**. For the FC task, XGBoost gain-based feature selection further reduces this to **60 features** (Repr. F), eliminating low-information channels while preserving the most discriminative ones. The top 5 features by XGBoost information gain are: E1 FFlow_min, OAT_min, amp2_mean, amp2_min, and volt2_min.

3.4 Representations G and H : Temporal Shape

Global statistics and inter-cylinder features cannot capture the *shape* of sensor trajectories: whether a temperature curve rises monotonically or fluctuates, whether RPM has a characteristic oscillation frequency, or whether fuel flow drops abruptly at a specific flight phase. To address this, two time-series transformation approaches are explored.

3.4.1 Representation G: MiniRocket

MiniRocket [4] transforms each univariate time series into a high-dimensional feature vector using a large bank of random convolutional kernels (default: 10,000) with dilated receptive fields. For each kernel applied to each of the 23 sensor channels:

$$f_k = \text{PPV}(\mathbf{x}_j * \mathbf{w}_k) = \frac{1}{T} \sum_t \mathbf{1}[(\mathbf{x}_j * \mathbf{w}_k)_t > 0] \quad (4)$$

where PPV (Proportion of Positive Values) is a simple pooling that is both fast to compute and highly discriminative. The resulting Repr. G vector has approximately 10,000 features per flight.

Repr. G₊₊ further concatenates the MiniRocket features with Repr. F₊₊, yielding ~10,224 features.

3.4.2 Representation H: tsfresh

tsfresh [5] computes a comprehensive library of time-series features per channel, including auto-correlation coefficients at multiple lags, FFT spectral peaks, entropy measures, nonlinearity statistics, and linear/nonlinear trend indicators. The feature count varies depending on the selected feature set but is on the order of hundreds to thousands per channel.

Table 6 summarizes all representations and their key properties.

Table 6: Summary of feature representations.

Repr.	Method	Dim.	What it captures
B	Global stats (8 per sensor)	184	Magnitude, spread, trend of full flight
B _{seg}	B applied to N segments	552–1840	Temporal evolution of global stats
F ₊₊	B + cross-cylinder features	224	Inter-cylinder asymmetry and ratios
F	F ₊₊ + XGB gain selection	60	Most discriminative 60 features
G	MiniRocket kernels	~10k	Temporal shape via random convolutions
H	tsfresh	~var	Autocorrelation, FFT, entropy, non-linearity
G ₊₊	G + F ₊₊	~10k+	Temporal shape + cross-physical

4. Methodology

4.1 Problem Formulation

Given a flight represented as a feature vector $\mathbf{f} \in \mathbb{R}^d$ (from any representation in Section 3), the system must produce one of 20 diagnostic labels:

$$\hat{y} \in \{0, 1, 2, \dots, 19\} \quad (5)$$

where label 0 denotes a *healthy* flight and labels 1–19 denote the 19 fault categories of Table 3. This 20-class problem is decomposed into two sequential sub-tasks:

1. **Anomaly Detection (AD):** Binary classification, $y_{AD} \in \{0 = \text{healthy}, 1 = \text{anomaly}\}$.
2. **Fault Classification (FC):** 19-class classification, $y_{FC} \in \{1, 2, \dots, 19\}$, performed *only* on flights flagged as anomalous by the AD stage.

4.2 Two-Stage Diagnosis Pipeline

The two-stage cascade architecture is motivated by the observation that AD and FC require fundamentally different inductive biases: AD benefits from global whole-flight context, while FC requires local fault-specific pattern resolution [3]. Jointly optimizing both objectives in a single model leads to a conflict where the global context needed for AD suppresses the local fault signatures needed for FC.

The pipeline routes each incoming flight through Stage 1 (AD). Only flights classified as anomalous proceed to Stage 2 (FC). Healthy flights receive label 0 and bypass the fault classifier entirely.

Figure 1 illustrates the architecture.

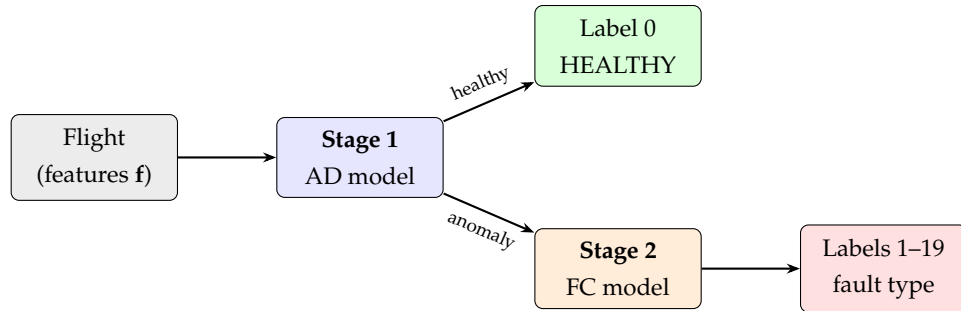


Figure 1: Two-stage diagnosis pipeline. Stage 1 (AD) routes flights to the healthy label or to Stage 2 (FC) for fault type identification.

4.3 Classical Models

Five classical scikit-learn models are evaluated on both the AD and FC tasks:

Logistic Regression (LR). A linear probabilistic classifier trained by maximizing the regularized log-likelihood. The decision boundary is a hyperplane in feature space: $\hat{p}(y = 1 | \mathbf{f}) = \sigma(\mathbf{w}^\top \mathbf{f} + b)$, where σ is the sigmoid function. LR is interpretable, fast to train, and robust to high-dimensional sparse features. The L2 regularization strength C is tuned per fold.

Linear Support Vector Machine (LinearSVM). Finds the maximum-margin hyperplane separating the two classes. Under class imbalance, the `class_weight='balanced'` option rescales the hinge loss by inverse class frequency. LinearSVM is equivalent to LR in the linear limit but optimizes the margin directly rather than the log-likelihood.

XGBoost (XGB). An ensemble of gradient-boosted decision trees [6]. XGBoost captures nonlinear feature interactions and implicitly handles feature importance through its information-gain-based splitting criterion. The `scale_pos_weight` parameter balances positive and negative class weights for AD; `class_weight='balanced'` is used for FC.

Random Forest (RF). A bagged ensemble of decision trees with feature subsampling at each split. RF provides implicit feature selection through out-of-bag importance scores and is robust to irrelevant features—important given that Repr. B contains 184 dimensions of which many are low-information for specific fault types.

K-Nearest Neighbors (KNN). Distance-based classifier that assigns the majority label among the k nearest neighbors in feature space. KNN serves as a sanity-check baseline; its performance is sensitive to the feature scale and to the curse of dimensionality in high-dimensional representations.

4.4 Deep Learning Benchmarks

Three deep learning architectures from the NGAFID benchmark paper [2, 3] serve as upper-bound comparisons. Unlike classical models, these operate directly on the raw 2,048-step time series, preserving all temporal resolution.

ConvTokMHSA. The ConvTok framework [2] first tokenizes the multivariate time series into non-overlapping patches using 1D CNN encoders that capture local shape descriptors. The tokens are then processed by a Multi-Head Self-Attention (MHSA) layer with full global receptive field—every token attends to every other token across the entire flight. This global aggregation is optimal for AD, where the overall health state is determined by the coherence of the whole-flight profile ($F1 = 0.799$). However, it suffers on FC because cross-stage operational noise dilutes weak local fault signatures ($F1 = 0.208$).

MMK Net. The Multi-Micro Kernel Network [3] employs parallel 1D convolutions with micro-scale kernels ($k = 1, 3, 5$) that explicitly restrict the receptive field to local temporal windows. This constraint prevents operational noise from dominating and isolates micro-pattern fault signatures. MMK Net deliberately omits

pooling layers to preserve temporal resolution, and uses LayerNorm instead of BatchNorm to stabilize training under severe class imbalance. It achieves the best FC performance ($F1 = 0.623$).

LMSD. The Long-Micro Scale Diagnostician [3] is the state-of-the-art end-to-end architecture that explicitly operationalizes the two-stage decomposition principle. It uses ConvTokMHSA (global receptive field) for Stage 1 AD and MMK Net (restricted receptive field) for Stage 2 FC, connected by a *hard-threshold routing mechanism* that enforces dimensional isolation: healthy samples trigger a label-0 output with the fault dimensions masked to $-\infty$, completely bypassing the FC head. Training is decoupled: the AD head trains on all 11,446 flights, while the FC head trains exclusively on the 5,601 anomalous flights, preventing the large healthy class from suppressing gradient signals for rare fault classes.

4.5 Oracle Hierarchical Classification

An additional experiment assesses the theoretical ceiling achievable through physics-based fault grouping. Ward hierarchical clustering is applied to the class-conditional feature centroids, grouping the 19 fault classes into K physically coherent super-groups. A separate XGBoost classifier is then trained for each group on the anomalous flights belonging to that group.

This *Oracle* configuration uses ground-truth group labels at test time (i.e., it knows which group a fault belongs to), providing an upper bound for the hierarchical approach. The $K = 4$ solution partitions the 19 classes as:

- **Group 0** (156 flights): Classes 10, 12 — cylinder structural failures
- **Group 1** (2,476 flights): Classes 6, 8, 14, 17 — engine operational issues
- **Group 2** (2,875 flights): Classes 0–5, 7, 9, 11, 15, 16, 18 — baffle/intake/engine maintenance
- **Group 3** (94 flights): Class 13 — overspeed events

4.6 Evaluation Metrics

4.6.1 F1-macro Score

The primary metric is the **macro-averaged F1 score**:

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K \frac{2P_k R_k}{P_k + R_k} \quad (6)$$

where P_k and R_k are precision and recall for class k , and K is the number of classes. Macro averaging treats all classes equally regardless of support, making it sensitive to performance on minority fault classes—which is precisely what matters for fault detection in long-tailed distributions.

4.6.2 Area Under the ROC Curve (AUC)

For AD, the AUC summarizes discrimination ability across all decision thresholds and is reported alongside F1. AUC is threshold-independent and provides a complementary view to F1 under varying operating points.

4.6.3 False Negative Rate (FNR)

The False Negative Rate measures the fraction of anomalous flights incorrectly classified as healthy:

$$\text{FNR} = \frac{FN}{FN + TP} \quad (7)$$

In an aviation safety context, a false negative—a missed fault—is far more consequential than a false positive (unnecessary inspection). FNR is therefore a primary safety metric.

4.6.4 MCWPM (Mission-Critical Weighted Performance Metric)

MCWPM is a safety-weighted F-score that penalizes false negatives more heavily than false positives:

$$\text{MCWPM} = \frac{\alpha \cdot R + P}{\alpha + 1} \quad (8)$$

where P is precision, R is recall, and $\alpha = 2.5$. Under this parameterization, a false negative contributes $\alpha^2 = 6.25$ times more to the loss than a false positive. MCWPM is reported alongside F1 and FNR for all AD experiments to provide a safety-contextualized performance assessment.

5. Results

5.1 Anomaly Detection

Table 7 reports 5-fold cross-validation performance for all classical models on Repr. B (184 features), the Ensemble variant on Repr. F₊₊ (224 features), and the published DL benchmarks.

Table 7: Anomaly Detection results (5-fold CV, mean $\pm$ std). Best classical result in **bold**; best DL result in *italics*.

Model	Repr.	F1	AUC	FNR	MCWPM
<i>Classical models (this work)</i>					
Logistic Reg.	B (184)	0.691$\pm$0.009	0.750$\pm$0.008	0.299$\pm$0.010	0.532$\pm$0.009
Linear SVM	B (184)	0.685 $\pm$ 0.008	0.748 $\pm$ 0.007	0.321 $\pm$ 0.008	0.522 $\pm$ 0.008
XGBoost	B (184)	0.682 $\pm$ 0.010	0.764 $\pm$ 0.013	0.332 $\pm$ 0.021	0.519 $\pm$ 0.022
Random Forest	B (184)	0.637 $\pm$ 0.011	0.722 $\pm$ 0.011	0.396 $\pm$ 0.024	0.473 $\pm$ 0.020
KNN	B (184)	0.531 $\pm$ 0.007	0.575 $\pm$ 0.004	0.482 $\pm$ 0.014	0.373 $\pm$ 0.008
Ensemble	F ₊₊ (224)	0.709$\pm$0.004	0.782$\pm$0.010	0.287$\pm$0.014	0.552$\pm$0.014
<i>Deep learning benchmarks (paper [2])</i>					
ConvTokMHSA	raw	0.799	—	0.163	—
InceptionTime [7]	raw	0.784	—	0.196	—
MMK Net	raw	0.721	—	0.298	—

5.1.1 Analysis

Best classical model. Logistic Regression with Repr. B achieves F1 = 0.691, AUC = 0.750, and FNR = 0.299. Its linear decision boundary over the 184-dimensional feature space is sufficient to capture the global health state of a flight. The simplicity of LR also makes it the most deployable classical option: fast inference, no hyperparameter sensitivity to training data, and full interpretability through its coefficient vector.

Ensemble improvement. Adding cross-physical inter-cylinder features (Repr. F₊₊) and combining LR with an XGBoost component raises F1 to 0.709 (+1.8 pp) and reduces FNR to 0.287 (−1.2 pp). The MCWPM improvement from 0.532 to 0.552 reflects a meaningful gain in safety-weighted performance. This confirms that inter-cylinder temperature asymmetry provides information beyond per-sensor global statistics.

Gap to DL. The best DL model (ConvTokMHSA, F1 = 0.799) outperforms the best classical system by **9.0 pp in F1** and **12.4 pp in FNR**. ConvTokMHSA applies global self-attention over the full 2,048-step sequence, allowing it to detect subtle whole-flight coherence patterns—such as gradual thermal drift correlated with electrical anomalies across flight phases—that are invisible to any fixed-window or global statistic. This gap represents the minimum cost of discarding temporal structure for AD.

Feature importance. Among the 184 Repr. B features, the five most discriminative sensors by classification effect size (Cohen’s d , healthy vs. anomalous) are: E1 OilP ($d = 0.158$), IAS ($d = 0.097$), E1 FFlow ($d = 0.087$), AltMSL ($d = 0.079$), and E1 RPM ($d = 0.075$). Oil pressure is the single most discriminative channel, consistent with its role as the primary engine health indicator and direct sensor for lubrication system faults.

5.2 Fault Classification

Table 8 reports FC results on the 5,601 anomalous flights across multiple representations and models, compared to DL benchmarks.

Table 8: Fault Classification results (5-fold CV, F1-macro). Best classical in **bold**.

Model	Repr.	F1 macro	Accuracy
<i>Classical — Repr. B baseline</i>			
XGBoost	B (184)	0.121±0.005	0.379
Logistic Reg.	B (184)	0.097±0.009	0.102
Linear SVM	B (184)	0.066±0.014	0.066
Random Forest	B (184)	0.034±0.005	0.375
<i>Classical — improved representations</i>			
XGBoost	F (60)	0.227±0.020	0.404
Oracle K = 4	F (60)	0.420±0.022	—
<i>Deep learning benchmarks (paper)</i>			
MMK Net [3]	raw	0.623	—
InceptionTime [7]	raw	0.563	—
ConvTokSWLA	raw	0.523	—
ConvTokMHSA	raw	0.208	—

5.2.1 Per-Class Analysis

Table 9 reports the per-class F1 scores for the best classical model (XGBoost, Repr. B) to illustrate the long-tail performance distribution.

Table 9: Per-class F1 (XGBoost, Repr. B). Classes sorted by sample count.

Fault Category	n	F1
Intake gasket leak / damage	2,097	0.533
Rocker cover leak / loose / damage	1,024	0.364
Baffle plug need repair	254	0.153
Baffle screw missing / loose	211	0.139
Oil cooler need maintenance	75	0.127
Engine seal / tube / bolt loose	76	0.098
Cylinder crack / fail / need repair	108	0.055
Engine run rough	141	0.051
Baffle tie / tie rod loose	144	0.047
Engine idle / RPM issue	93	0.038
Pilot / in-flight noticed issue	75	0.000

5.2.2 Analysis

Repr. B baseline. XGBoost with Repr. B achieves $F1 = 0.121$, classifying with reasonable accuracy only the two head classes (intake gasket and rocker cover), which together account for 55.7% of anomalous flights. Minority classes are largely misclassified as the dominant class—a direct consequence of both the long-tailed distribution and the global statistics discarding the local fault signatures.

Feature selection gain. XGB gain-based selection to 60 features (Repr. F) nearly *doubles* $F1$ from 0.121 to 0.227 (+88%). This improvement demonstrates that the 124 features removed from Repr. B were actively harmful—they added noise that confused the classifier—rather than simply uninformative. The 60 retained features are dominated by electrical channels (`amp2`, `volt2`) and fuel flow extrema (`E1_FFlow_min`), highlighting the diagnostic relevance of transient low values in these channels.

Oracle hierarchical ceiling. Grouping the 19 classes into 4 physically coherent groups via Ward clustering and training a per-group XGBoost raises $F1$ to 0.420 (+96% over flat XGB with Repr. F). This result quantifies the theoretical gain from perfect fault taxonomy knowledge: if the type of fault were known at the coarse level (engine operational vs. baffle/intake maintenance), the intra-group classification becomes substantially easier because groups share similar sensor signatures.

Structural gap to DL. Despite the Oracle improvement to 0.420, the best DL model (MMK Net, $F1 = 0.623$) still exceeds the classical ceiling by **20 pp**. This residual gap is structural: MMK Net’s micro-kernel convolutions ($k = 1, 3, 5$) with no pooling extract local temporal signatures at the time step level, while even the Oracle hierarchical approach is constrained by the information available in flight-level global statistics.

Remarkably, ConvTokMHSA (global attention, $F1 = 0.208$) performs *worse* than flat XGBoost with Repr. F (0.227), confirming that whole-flight context *hurts* FC by injecting operational noise into the attention weights.

5.3 End-to-End Diagnosis

5.3.1 Performance Ladder

Table 10 reports the full E2E performance ladder as a function of increasing oracle knowledge.

Table 10: E2E performance ladder (F1-macro, 20 classes: 0 = healthy, 1–19 = fault type). The “oracle” prefix denotes perfect Stage 1 or Stage 2 routing.

Configuration	F1 macro	Δ vs. Cascade
Naive baseline (all HEALTHY)	0.034	—
Cascade LR→XGB (Repr. B)	0.112	—
Cascade XGB (Repr. F)	0.157	+0.045
Oracle routing → XGB (Repr. B)	0.165	+0.053
Var. A: real AD + Oracle FC ($K = 4$)	0.342	+0.230
Var. B: Oracle AD + Oracle FC ($K = 4$) [ML ceiling]	0.457	+0.345
LMSD (ConvTokMHSA + MMK Net) [3]	0.623	—

5.3.2 Error Decomposition

For the Cascade LR→XGB pipeline on the 11,446 out-of-fold flights, Table 11 decomposes the total errors by type.

Table 11: E2E error decomposition for Cascade LR→XGB (Repr. B), out-of-fold.

Error type	Count	% of all flights
Correct prediction	5,538	48.4%
Stage 1 false negative (fault → healthy)	1,675	14.6%
Stage 1 false positive (healthy → fault)	1,833	16.0%
Stage 2 wrong fault type	2,400	21.0%
Total errors	5,908	51.6%

5.3.3 Gap Analysis

Three quantities characterize the performance gap relative to the DL upper bound:

$$\Delta_{\text{routing}} = F1(\text{Var. A}) - F1(\text{Cascade}) = 0.342 - 0.112 = +0.230 \quad (9)$$

$$\Delta_{\text{AD}} = F1(\text{Var. B}) - F1(\text{Var. A}) = 0.457 - 0.342 = +0.115 \quad (10)$$

$$\Delta_{\text{residual}} = F1(\text{DL}) - F1(\text{ML ceiling}) = 0.623 - 0.457 = -0.166 \quad (11)$$

These results reveal three important findings:

1. **FC is the dominant bottleneck.** The routing gain $\Delta_{\text{routing}} = +0.230$ (Equation 9) dwarfs the AD gain $\Delta_{\text{AD}} = +0.115$. Improving fault classification has twice the leverage of improving anomaly detection for E2E performance.
2. **Stage 1 false negatives are the safety-critical failure mode.** Of the 5,908 total errors, 1,675 (28.4%) are Stage 1 false negatives—faulty flights routed to the healthy label. These never reach the fault classifier and are the most dangerous error type in an operational context.
3. **40% of errors are Stage 2 wrong-type errors.** Stage 2 errors (2,400, 40.6% of all errors) indicate correct anomaly detection followed by incorrect fault attribution. This is not a safety failure (the flight is still flagged), but prevents actionable maintenance recommendations. The 0.166 residual gap between the ML ceiling and LMSD is irrecoverable with global statistical features; only temporal micro-pattern features (as in MMK Net) can close it.

5.4 Summary Comparison

Table 12 consolidates the best results across all tasks and compares them against DL benchmarks.

Table 12: Final performance summary. $\uparrow$ = higher is better; $\downarrow$ = lower is better.

Task	Best model / method	F1 $\uparrow$	FNR $\downarrow$	vs. DL best
AD	LR Repr. B	0.691	0.299	−0.108
AD	Ensemble Repr. F ₊₊	0.709	0.287	−0.090
AD	ConvTokMHSA (DL)	0.799	0.163	—
FC	XGB Repr. B (baseline)	0.121	—	−0.502
FC	XGB Repr. F (selected)	0.227	—	−0.396
FC	Oracle K = 4 (ceiling)	0.420	—	−0.203
FC	MMK Net (DL)	0.623	—	—
E2E	Cascade LR→XGB Repr. B	0.112	—	−0.511
E2E	Cascade XGB Repr. F	0.157	—	−0.466
E2E	Var. B ML ceiling	0.457	—	−0.166
E2E	LMSD (DL)	0.623	—	—

The results confirm the hypothesis stated in Section 1: classical models with global statistical features are *competitive but not equivalent* to DL for AD (−9 pp gap), and face a *structural and irrecoverable* gap for FC (−20 pp gap even at the Oracle ceiling). The path to closing the remaining FC gap lies in adopting temporal deep learning models (specifically, locally restricted architectures like MMK Net) rather than further feature engineering on global statistics.

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